

AI-Enhanced Learning Insights & Student Support

AI-POWERED LEARNING, ALL IN ONE PLACE

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22 April - 8 August

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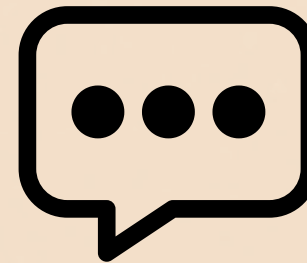
Introduction & Objectives

Problems Identified

Students



**Poor Visibility into
Academic Performance**



**Lack of Personalised
Feedback**



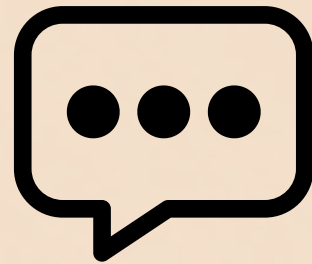
**Inefficient Monitoring of
Student Performance**

Project Objectives

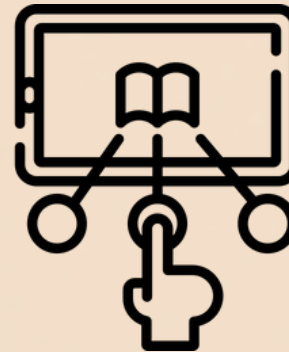
- 1** Leverage GenAI for real-time insights, personalised feedback, learning/revision materials
- 2** Design & develop a unified platform which includes student engagement & performance data
- 3** Develop a modular, scalable system architecture with the capability to integrate future systems & AI advancements

Solution

Students



**Personalised feedback
based on individual data**



**Interactive learning
with bite-sized media**

Educators



**Efficient Class-Level &
Individual Performance
Analysis**

System Architecture

Internal student systems



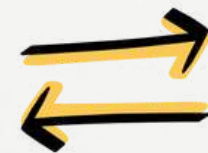
ETL pipeline



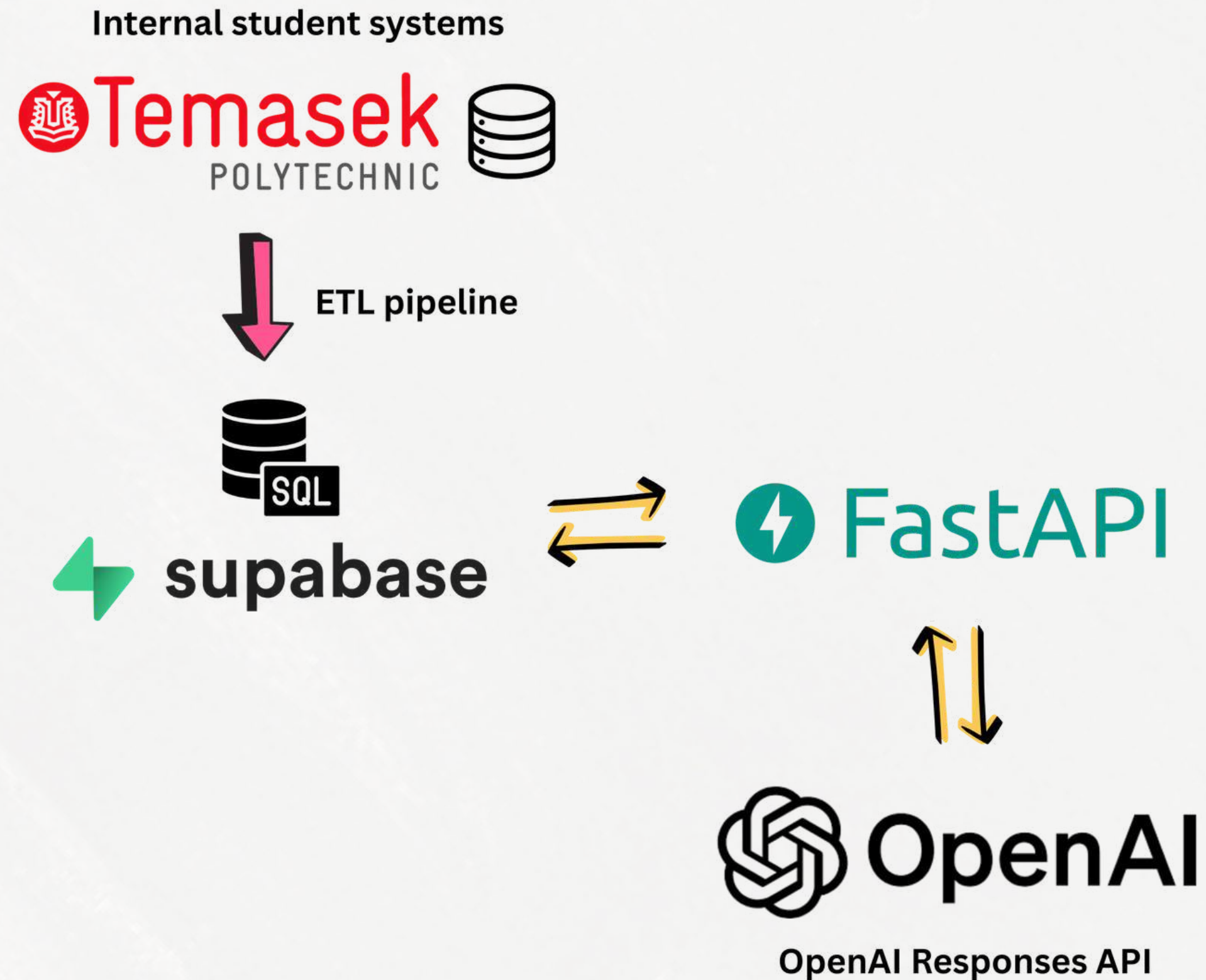
supabase

System Architecture

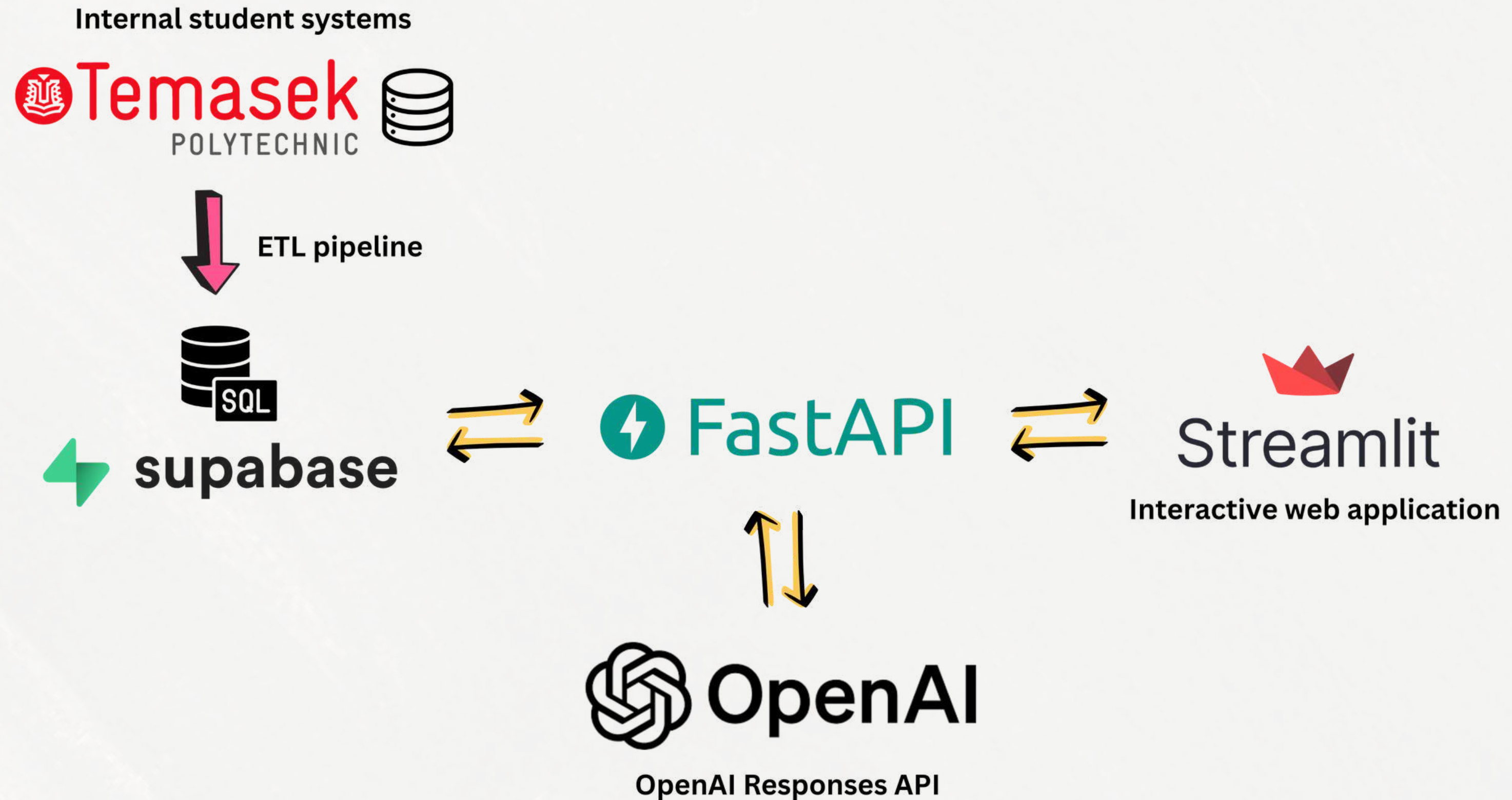
Internal student systems



System Architecture

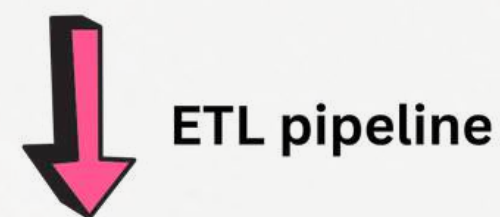


System Architecture



System Architecture

Internal student systems
 **Temasek**
POLYTECHNIC 



 **supabase**

 **+ a b | e a u**

Data visualization



 **FastAPI**



 **Streamlit**

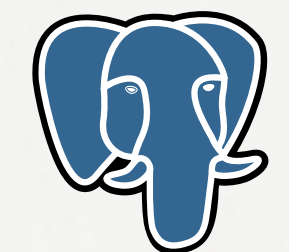
Interactive web application



 **OpenAI**

OpenAI Responses API

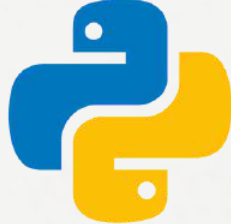
Languages used:



PostgreSQL

Solutions & Findings

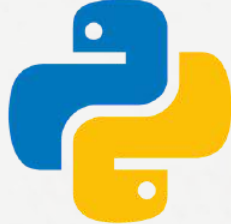
01 Synthetic Data Generation

Uses:   OpenAI

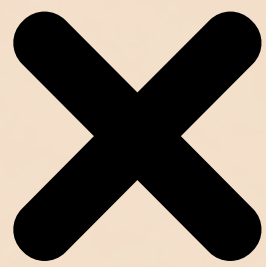
- 1 Define schema
- 2 Design prompt → Check generated data
- 3 Send prompts to OpenAI API in parallel
→ Obtain results

```
68 result, status = create_chat_completion(prompt, model=model)
69
70 return result, status
71
72 def generate_single_question(quiz_id: str, module_name: str, qnum: int):
73     prompt = (
74         f"You are creating question QN{qnum} for a quiz on the topic {module_name}."
75         f"Each topic has multiple sub-topics. Ensure that the question is relevant to the topic."
76         "Ensure your responses are strictly in the English language."
77         "Try to be as creative as possible in creating a question."
78         "Output EXACTLY ONE QUESTION IN THIS FORMAT:\n"
79         "<question text>"
80     )
81     # temperature = 2 gives more variation in output, but need to be careful
82     # else temperature = 1.5 is generally very stable
83     text, status = run_model(prompt, model='mini', temperature=1.5)
84     if status != "Success":
85         raise RuntimeError(f"API error for {quiz_id} QN{qnum}: {status}")
86     question_text = text.strip()
87     return {"quiz_id": quiz_id, "question_text": question_text}
88
89 # Assume quiz_data is a DataFrame already loaded in your environment
90 # quiz_data = pd.read_csv("quiz_data.csv")
91
92 all_questions = []
93 global_counter = 1
94
95 # Avoid race conditions
96 # don't take a big lock
```

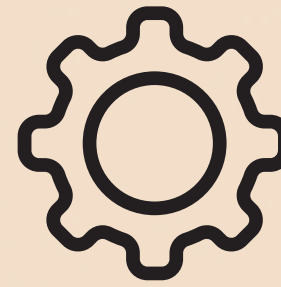
01 Synthetic Data Generation

Uses:   OpenAI

Benefits



**Doesn't require real
student data**



**Highly customizable
through prompting**



**Highly scalable
(within API rate limits)**

02 Backend-as-a-Service

Uses: ⚡ supabase



- 1 Set up managed PostgreSQL database (e.g. enabling vector extensions)
- 2 Load in generated data via Supabase API
- 3 Developing SQL functions & SQL views to serve data to the front-end



02 Backend-as-a-Service

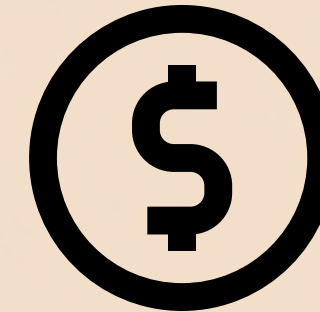
Uses:  supabase



Benefits



**Managed PostgreSQL
database & services**



**Scalability & cost-
effectiveness (this
project uses Free tier)**

03 Streamlit Web Framework

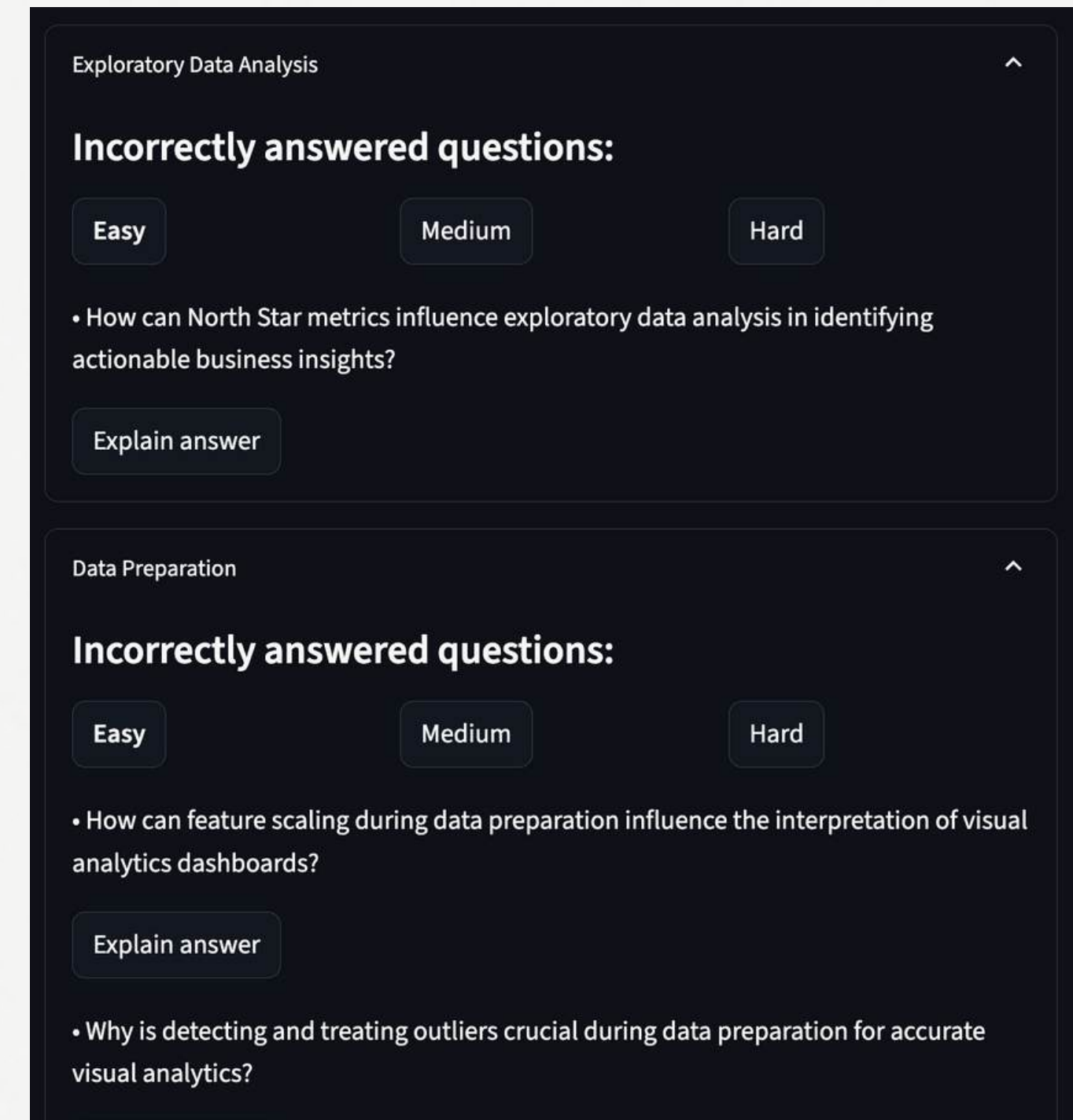
Uses:  Streamlit 

1

Use Python to write the main UI components

2

Custom HTML, CSS, & JavaScript for features and UI elements



03 Streamlit Web Framework

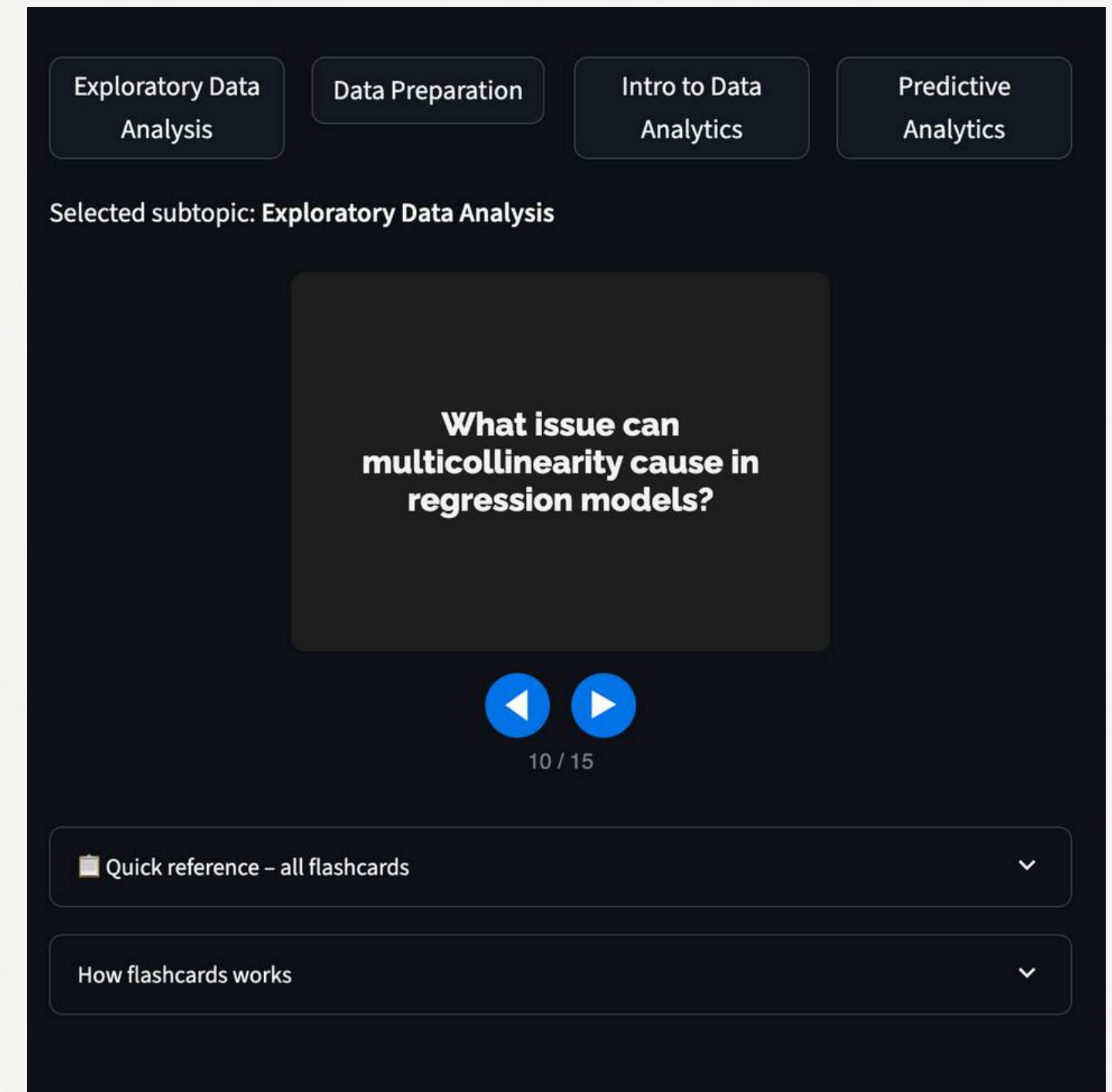
Uses:  Streamlit 

1

Use Python to write the main UI components

2

Custom HTML, CSS, & JavaScript for custom features and custom UI elements



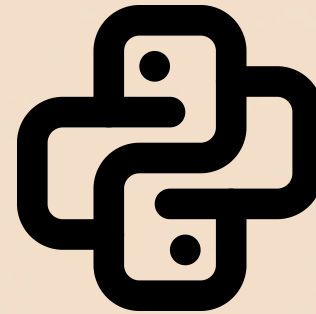
03 Streamlit Web Framework

Uses:  Streamlit 

Benefits



**Built for data science /
ML web apps**

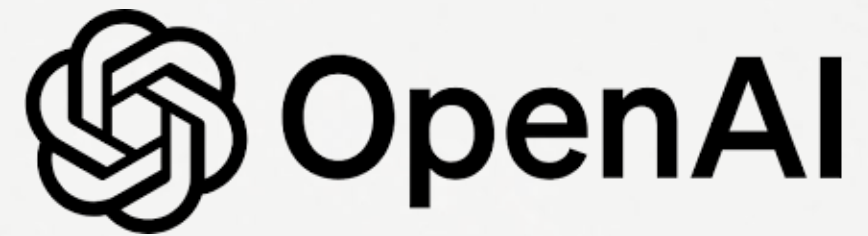


**Python-first design with
built-in UI components**



**Easy to deploy to
Streamlit Cloud / cloud
containers**

04 Generative AI features



Responses API

GPT 4.1 mini

GPT 5 chat

GPT 4o mini TTS

Whisper (speech-to-text)

04 Generative AI features

Uses:  OpenAI

For students

Module summary

Exploration of wrong answers

Revision flashcards

Interactive learning

Bite-sized media content



04 Generative AI features

Uses:  OpenAI

For educators

Detailed descriptive analytics

Detailed prescriptive interventions

Common mistakes

Analysis of individual low-performers

Email reporting



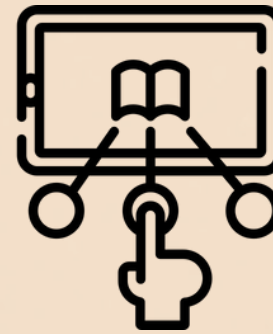
04 Generative AI features

Uses:  OpenAI

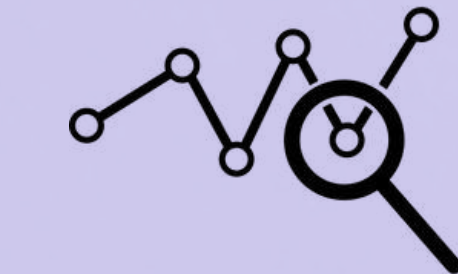
Students



**Personalized learner
feedback**



**Interactive, multi-modal
learning experience**



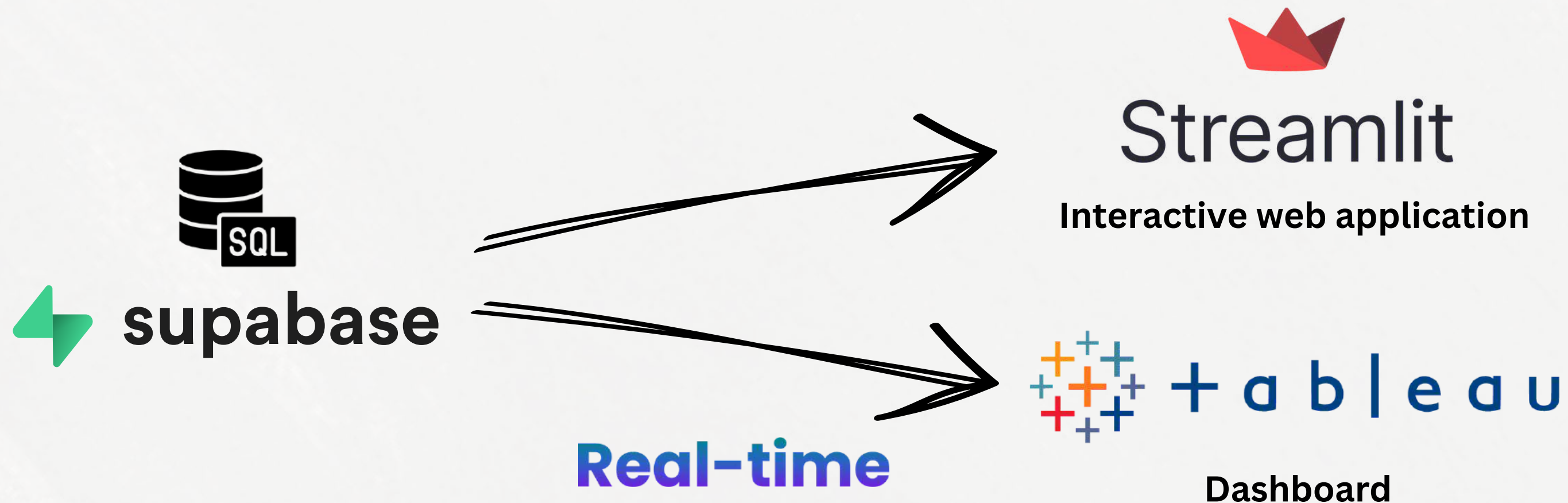
**Effective, automated
descriptive &
prescriptive analytics**

05 Data Visualization

Uses:  + a b l e a u

Student & teacher dashboards on Tableau
(integrated into web app)

Built-in charts



06 Unsupervised Clustering

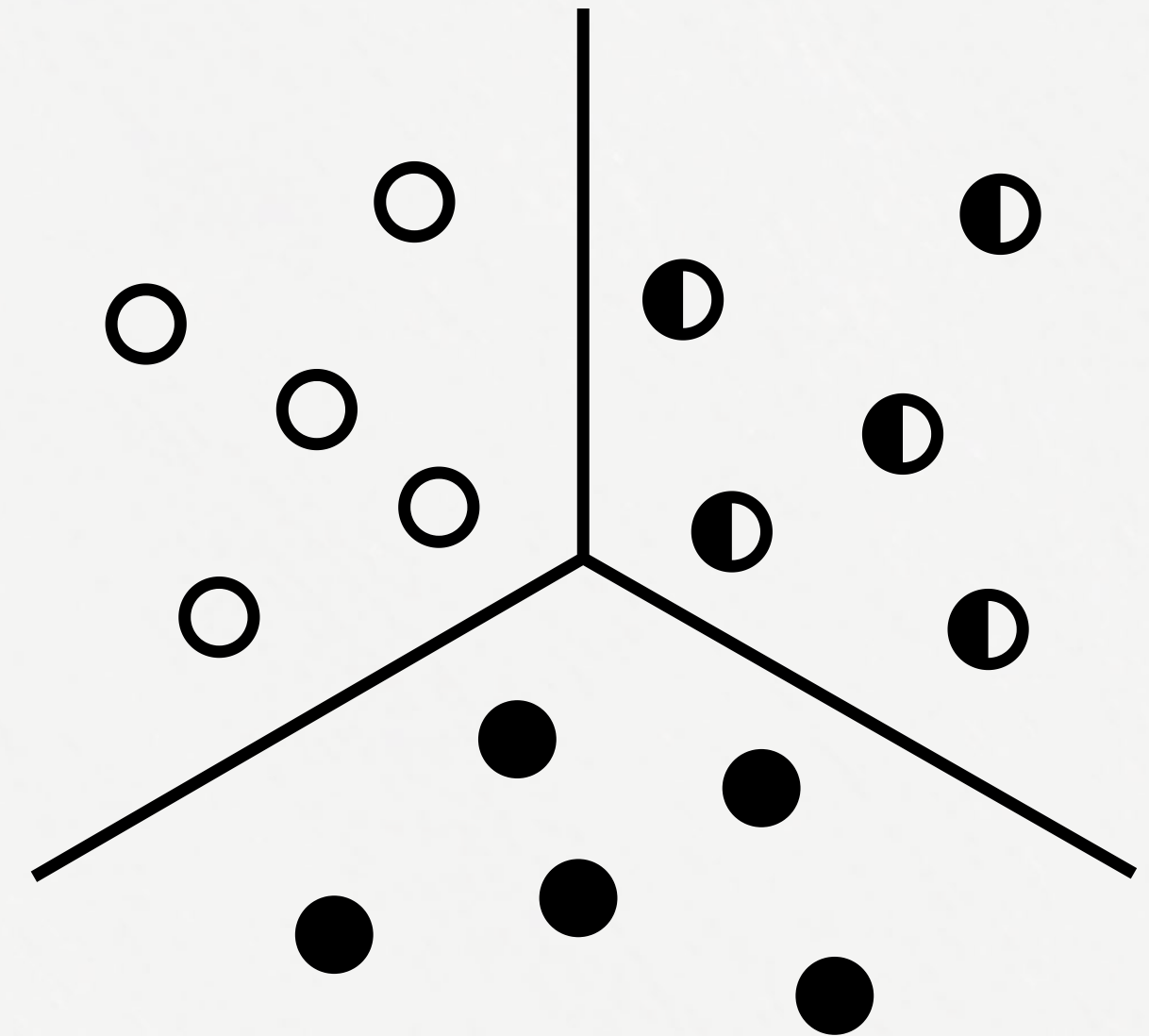
Uses:



Utilises k-means clustering (k=4)

Segment students by results

Discover patterns →
targeted intervention strategies



06 Unsupervised Clustering

Uses:

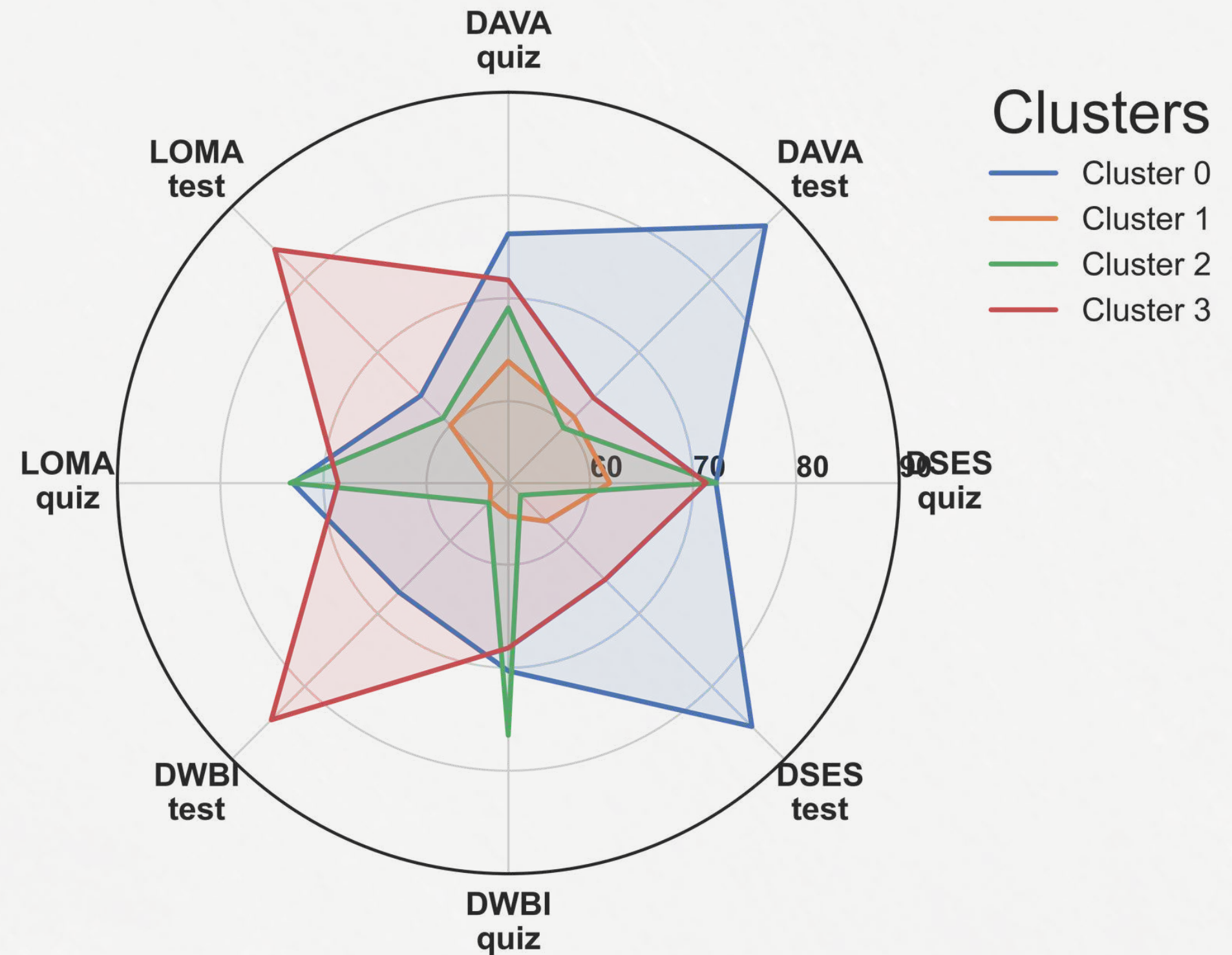


Cluster 0:
Strong at DAVA & DSES
Weaker at LOMA & DWBI

Cluster 1:
Weak in all subjects

Cluster 2:
Decent at quizzes, weak at tests
across all subjects

Cluster 3:
Strong at LOMA & DWBI
Weaker at DAVA & DSES



Scalability & Business Impact

Business Impact: Students

Learner

- Personalized learning materials
- Engaging learning environment with written, spoken, visual media

Data

- Integrates with assessment results, student engagement, and more
- Tailors content to each student

Analytics

- Uses charts, tables, summaries to let students explore their results
- Highlight each student's strengths and weaknesses

Intervention

- Personalized feedback & explanations for mistakes
- Revision flashcards & videos based on past mistakes

Business Impact: Educators

Learner

- Enables data-driven lesson plans

Data

- Unified platform for data exploration & analysis
- No need for manual data collection

Analytics

- Identify weak students & common mistakes
- Enable class-level or individual analysis with just a few clicks

Intervention

- Provide class-level & individual targeted feedback
- Automated email reporting
- Promotes self-learning → reduced educator workload

Scalability & Adaptability

Supabase backend

- Flexible scaling with affordable pricing tiers
- No manual deployment effort

Streamlit

- Streamlit Cloud deployment
- Easily containerized and deployed to cloud containers (e.g. AWS Fargate)

Generative AI

- Flexible and easy to migrate to newer Gen AI models, or models from a different API
- Just need to modify a few lines of code to change the model

Future Enhancements

Future Enhancements

More powerful RAG system

- Current RAG system only applies **basic RAG techniques** against **educator-provided** learning materials
- Can implement **multi-vector retriever**, **summary embedding** and **fine-tuning** for better search and retrieval
- Allow students to **upload own notes and materials** for more personalisation

Wider variety of interactive learning tools

- **AI podcast**: generates podcast based on learning materials
- **AI quizzes**: generates unlimited quizzes based on learning materials for unlimited revision
- **AI notes platform**: lets students search, query, and summarise their own notes

All of these features have been implemented in some form by other software tools, such as Google NotebookLM, TurboLearn AI, and more (TurboLearn AI is a paid AI-powered learning platform for \$30/month!)

Lessons Learnt

Lessons Learnt

Soft Skills:

Time Management & Planning

- Juggle **MP** with my **electives**
- **Multiple submissions** in a week
- Learnt to **carefully plan and manage my time** to complete the project and my assignments **on time**

Technical Skills:

New Technical Tools

- MP provided an opportunity for me to **learn new tools** such as Supabase
- I learned how to utilise **generative AI models** to build a variety of tools and services



*Thank
you!*